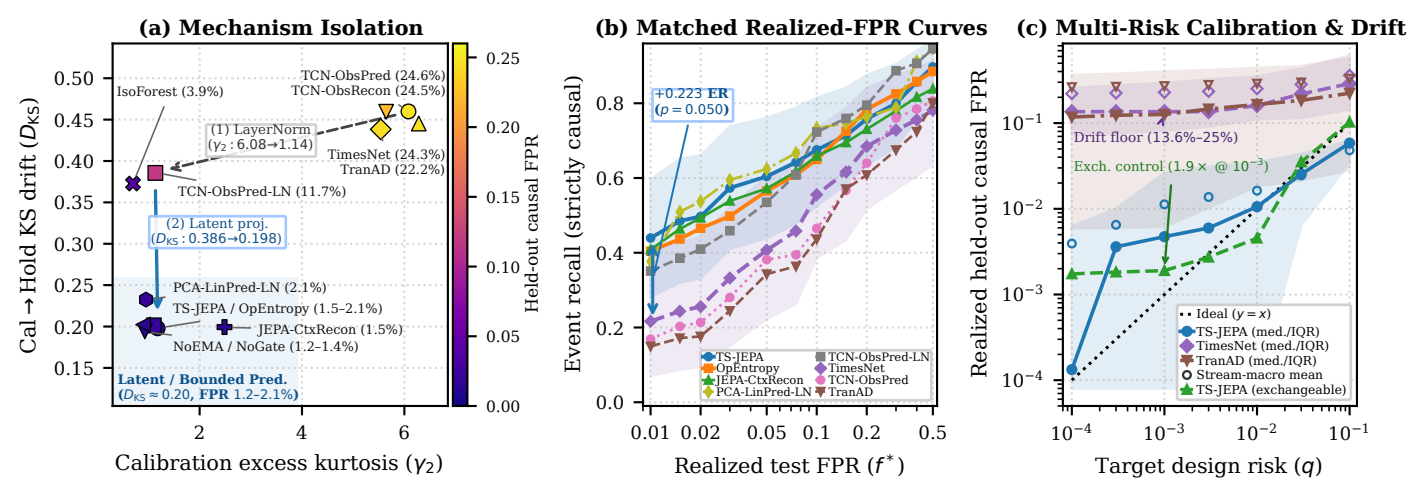

(a) Mechanism Isolation
Cal → Hold KS drift (D_{KS})
Calibration excess kurtosis (γ_2)
Held-out causal FPR
IsoForest (3.9%)
TCN-ObsPred (24.6%)
TCN-ObsRecon (24.5%)
(1) LayerNorm (γ_2 : 6.08→1.14)
TimesNet (24.3%)
TranAD (22.2%)
TCN-ObsPred-LN (11.7%)
(2) Latent proj. (D_{KS} : 0.386→0.198)
PCA-LinPred-LN (2.1%)
TS-JEPA / OpEntropy (1.5–2.1%)
JEPA-CtxRecon (1.5%)
NoEMA / NoGate (1.2–1.4%)
Latent / Bounded Pred.
($D_{KS} \approx 0.20$, FPR 1.2–2.1%)
(b) Matched Realized-FPR Curves
Event recall (strictly causal)
Realized test FPR (f^*)
+0.223 ER
($p = 0.050$)
TS-JEPA
OpEntropy
JEPA-CtxRecon
PCA-LinPred-LN
TCN-ObsPred-LN
TimesNet
TCN-ObsPred
TranAD
(c) Multi-Risk Calibration & Drift
Realized held-out causal FPR
Target design risk (q)
Drift floor (13.6%–25%)
Exch. control (1.9 × @ 10^{-3})
Ideal ($y = x$)
TS-JEPA (med./IQR)
TimesNet (med./IQR)
TranAD (med./IQR)
Stream-macro mean
TS-JEPA (exchangeable)